

Optimization

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Outline

Motivation

Entropy

Conditional Entropy and Mutual Information

Cross-Entropy and KL-Divergence



Let's work on this subject in our Optimization lecture

Cross Entropy

Cross Entropy: The expected number of bits when a wrong distribution Q is assumed while the data actually follows a distribution P

Objective function

$$H(p, q) = - \sum_{x \in \mathcal{X}} p(x) \log q(x) = \underbrace{H(P)}_{\text{actual pdf}} + \underbrace{KL[P][Q]}_{\text{predicted}}$$

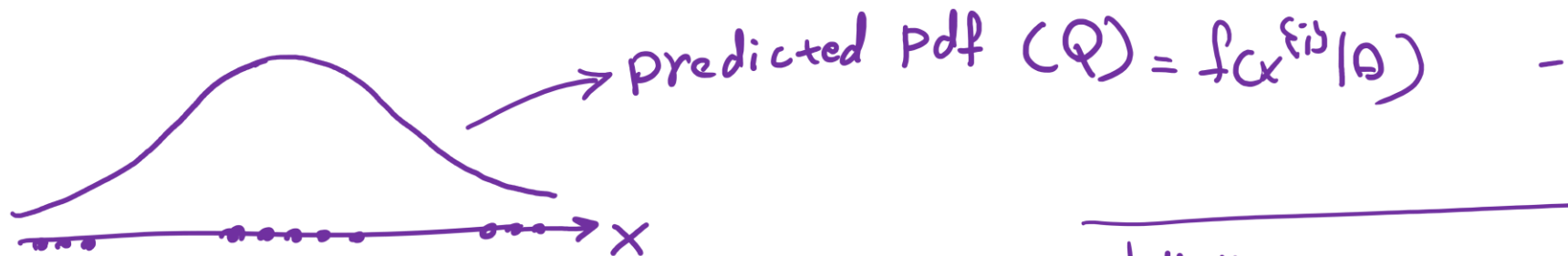
loss function

This is because:

$$H(p, q) = \mathbb{E}_p[l_i] = \mathbb{E}_p \left[\log \frac{1}{q(x_i)} \right]$$

$$H(p, q) = \sum_{x_i} p(x_i) \log \frac{1}{q(x_i)}$$

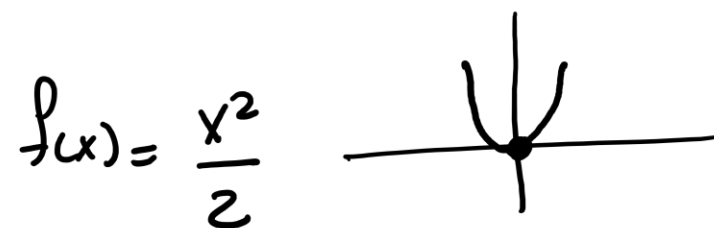
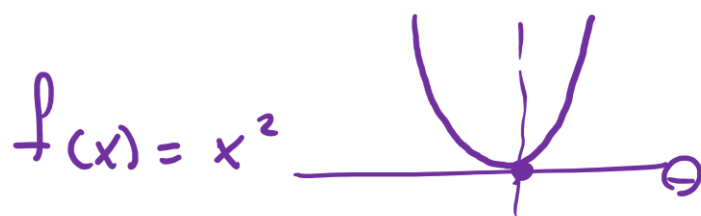
$$H(p, q) = - \sum_x p(x) \log q(x).$$



$$L(\theta | x) = \prod_{i=1}^N f(x^{i3} | \theta)$$

Objective function

$$\boxed{\text{Max } l(\theta | x) = \sum_{i=1}^N \log f(x^{i3} | \theta)}$$



Min negative log likelihood

$$\frac{1}{N} \sum (\cdot)^{g(x)} = E[\cdot]^{g(x)}$$

GLR

$$\text{Min } l(\theta | x) = - \sum_{i=1}^N \log f(x^{i3} | \theta)$$

$$\text{Min } l(\theta | x) = \frac{1}{N} \sum \left(- \log f(x^{i3} | \theta) \right) = E \left[- \log f(x^{i3} | \theta) \right]$$

avg negative log likelihood = CE

$$= E \left[- \log Q \right]$$

$$E[-\log P + \log P - \log Q]$$

$$\log P - \log Q = \log \frac{P}{Q} \\ = -\log \frac{Q}{P}$$

$$E[A+B+C] = E[A] + E[B+C]$$

$$E[g(x)] = \sum \underline{p(x)} \underline{g(x)}$$

$$E[-\log P] + E[-\log \frac{Q}{P}]$$

↓

$$\sum p(x) * (-\log p(x)) + \sum p(x) * (-\log \frac{Q(x)}{p(x)})$$

$$\left(\sum p(x) \log_2 \frac{1}{p(x)} \right) - \sum p(x) \log \frac{Q(x)}{p(x)}$$

$$H(P) + KL[P][Q] = CE = \frac{1}{N} \sum -\log f(x^{(i)} | \theta)$$

$$-\cancel{\sum p(x) \log p(x)} - \sum p(x) \log Q(x) + \cancel{\sum p(x) \log p(x)}$$

$$-\sum p(x) \log Q(x) = CE$$

Labeling target values

Label encoding (ordinal) and One-hot encoding

$$X = \begin{bmatrix} \text{height} & \text{weight} & \text{age} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{n \times d} \rightarrow Y_a = \begin{bmatrix} \text{fish} \\ \text{cat} \\ \text{dog} \\ \vdots \end{bmatrix}_{n \times 1} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \end{bmatrix} \rightsquigarrow \text{ML} \rightsquigarrow \hat{Y}_p = \begin{bmatrix} 0.5 \\ 1.2 \\ 1.8 \\ \vdots \end{bmatrix}$$

$$\text{fish} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \quad \text{cat} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \quad \text{dog} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$Y_a = \begin{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ \vdots \end{bmatrix} \rightsquigarrow \text{ML} \rightsquigarrow \hat{Y}_p = \begin{bmatrix} \begin{bmatrix} 78 & 20 & 10 \end{bmatrix} \\ \begin{bmatrix} 2 & 92 & 3 \end{bmatrix} \\ \begin{bmatrix} 2 & 3 & 80 \end{bmatrix} \\ \vdots \end{bmatrix} \rightsquigarrow \text{Softmax} \rightsquigarrow \begin{bmatrix} \begin{bmatrix} 0.8 & 0.1 & 0.1 \end{bmatrix} \\ \begin{bmatrix} 0.2 & 0.7 & 0.1 \end{bmatrix} \\ \begin{bmatrix} 0.3 & 0.1 & 0.6 \end{bmatrix} \\ \vdots \end{bmatrix}$$

$$\text{Max} \sum_{i=1}^N Y_a^{(i)} \cdot \hat{Y}_p^{(i)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.1 \\ 0.1 \end{bmatrix} + \dots + \dots \Rightarrow \text{maximum possible is } = N$$

$$\text{Min} \sum_{i=1}^N \| Y_a^{(i)} - \hat{Y}_p^{(i)} \|_2^2 = \left[(1-0.8)^2 + (0-0.1)^2 + (0-0.1)^2 + \dots + \dots \right]$$

Why Cross entropy and not simply use dot product?

$\mathbb{R} \in$

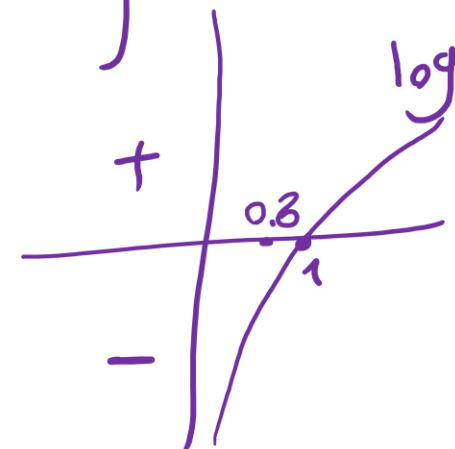
$$CE = - \sum_{i=1}^N y_a^{(i)} \log \hat{y}_p^{(i)}$$

$$y_a = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & & \end{bmatrix}$$

$$\hat{y}_p = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ \vdots & & \end{bmatrix}$$

$$CE = - \left(\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \log 0.8 \\ \log 0.1 \\ \log 0.1 \end{bmatrix} + \dots \right)$$

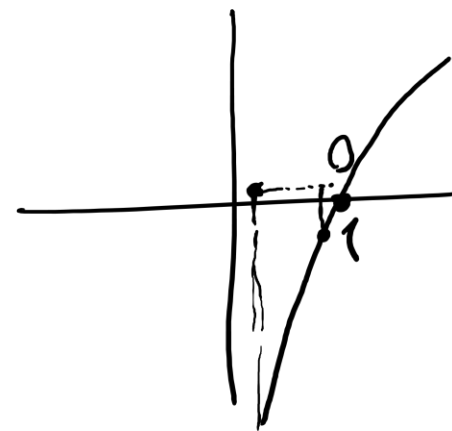
$y_a^{(i)}$ $\hat{y}_p^{(i)}$



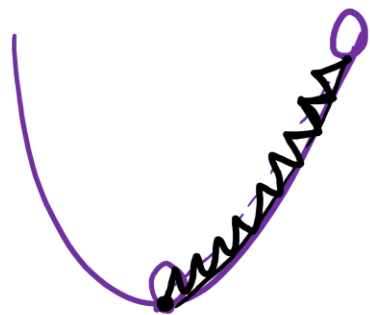
① $-\sum y_a^{(i)} \cdot \hat{y}_p^{(i)}$



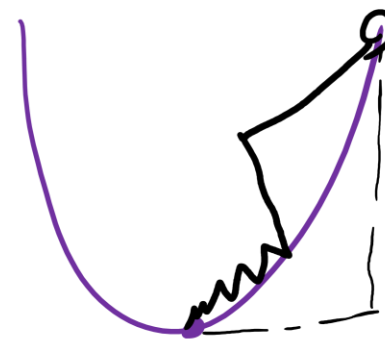
② $-\sum y_a^{(i)} \log \hat{y}_p^{(i)}$



$f(x) = x^2$
 $f'(x) = 2x = 0$



$f(x) = x^2$



Kullback-Leibler Divergence

Another useful information theoretic quantity measures the difference between two distributions.

$$\begin{aligned}\mathbf{KL}[P(S)||Q(S)] &= \sum_s P(s) \log \frac{P(s)}{Q(s)} \\ &= \underbrace{\sum_s P(s) \log \frac{1}{Q(s)}}_{\text{cross entropy}} - \mathbf{H}[P] = H(P, Q) - H(P)\end{aligned}$$

Excess cost in bits paid by encoding according to Q instead of P .

KL Divergence is
a **KIND OF**
distance
measurement

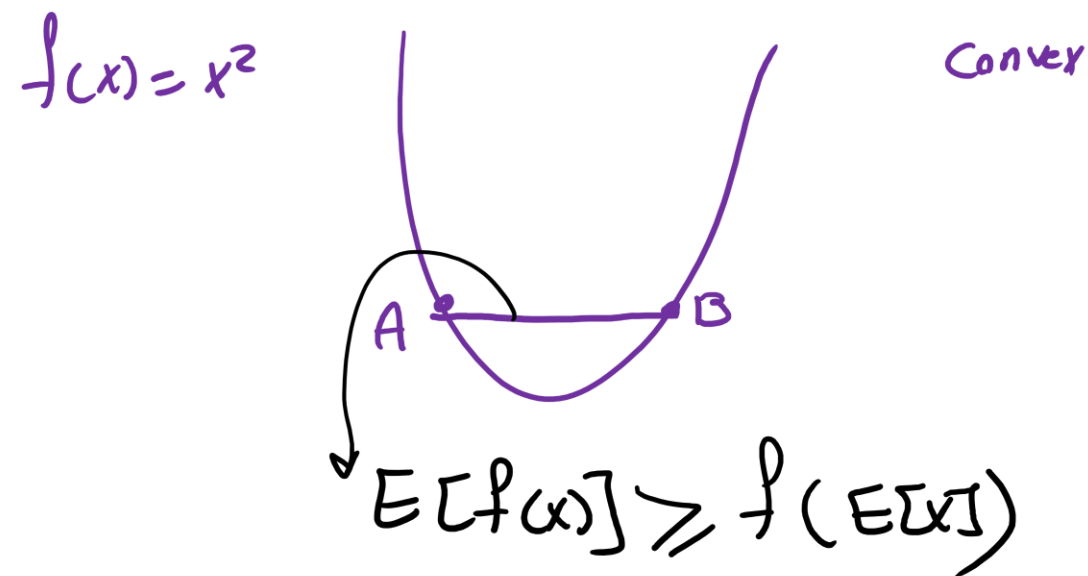
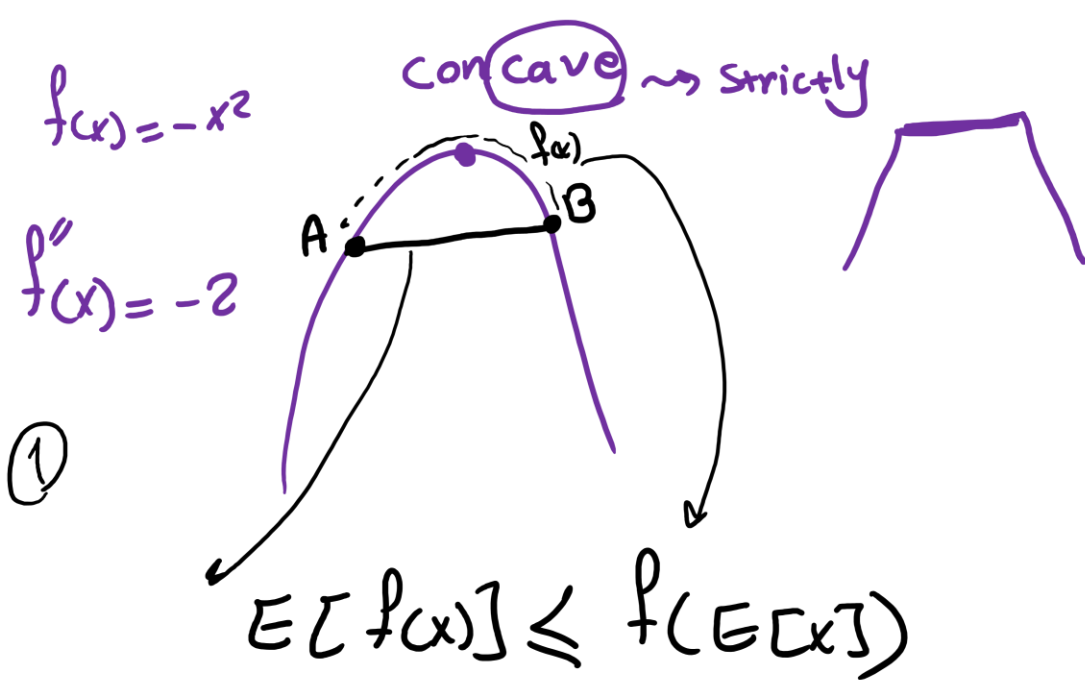
$$-\mathbf{KL}[P||Q] = \sum_s P(s) \log \frac{Q(s)}{P(s)}$$

log function is
concave or
convex?

$$\begin{aligned}\sum_s P(s) \log \frac{Q(s)}{P(s)} &\leq \log \sum_s P(s) \frac{Q(s)}{P(s)} && \text{By Jensen Inequality} \\ &= \log \sum_s Q(s) = \log 1 = 0\end{aligned}$$

So $\mathbf{KL}[P||Q] \geq 0$. Equality iff $P = Q$

When $P = Q$, $KL[P||Q] = 0$

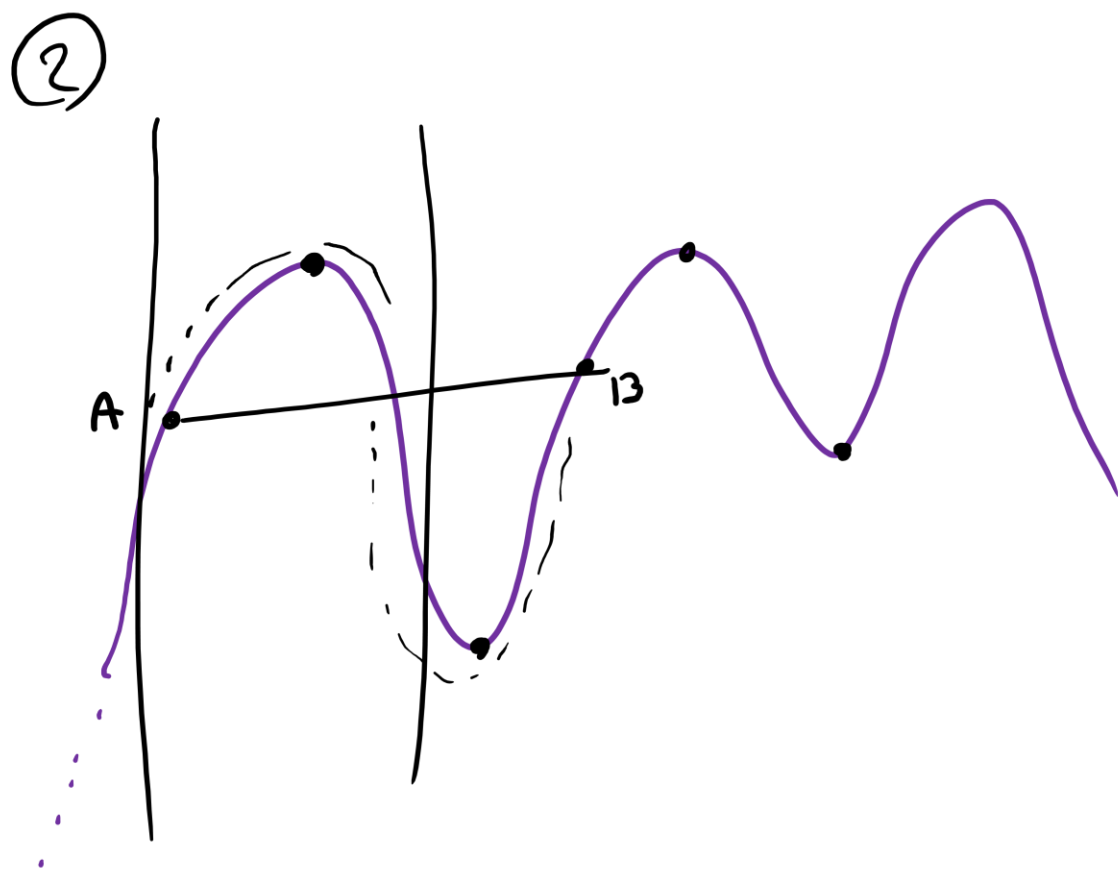


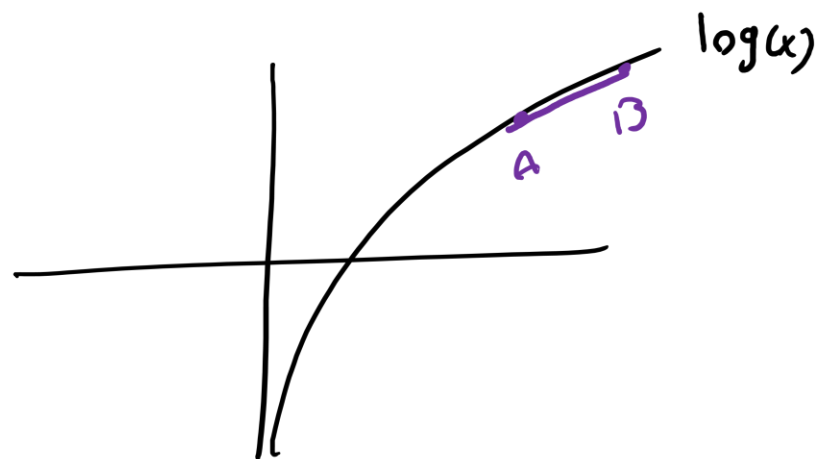
$f(x) = x^2$ $x = [1, 2]$
 $P(x) = [\frac{1}{2}, \frac{1}{2}]$

$E[f(x)] = \sum P(x) f(x) =$
 $P(x=1) f(x=1) + P(x=2) f(x=2)$
 $\frac{1}{2} * 1 + \frac{1}{2} * 4 = 2.5$

$f(E[x]) = f(E[x]=1.5) = (1.5)^2$

$E[x] = \sum P(x) * x = \frac{1}{2} * 1 + \frac{1}{2} * 2 = 1.5$





$$E[f(x)] \leq f(E[x]) \Rightarrow E[\log x] \leq \log(E[x])$$

$$f(x) = \log x$$

$$-KL[P][Q] = \sum p(x) \log \left(\frac{q(x)}{p(x)} \right) = \sum p(x) \log q(x)$$

$$E[\log q(x)] \leq \log E[q(x)]$$

$$\begin{aligned} &\leq \log \sum p(x) q(x) \\ &\leq \log \sum \cancel{p(x)} \frac{q(x)}{\cancel{p(x)}} \\ &\leq \log \left(\sum q(x) \right) \end{aligned}$$

$$-KL[P][Q] \leq \log 1 = 0 \Rightarrow KL[P][Q] \geq 0$$

Objective function $\leftarrow f(x, y)$

S.t. $g(x, y) = 0 \rightsquigarrow$ Equality constraint

S.t. $h(x, y) \leq 0 \rightsquigarrow$ Inequality constraint (KKT conditions)

$$f''(x, y) \rightsquigarrow H = \begin{bmatrix} \frac{\partial^2 f(x, y)}{\partial x^2} & \frac{\partial^2 f(x, y)}{\partial x \partial y} \\ \frac{\partial^2 f(x, y)}{\partial y \partial x} & \frac{\partial^2 f(x, y)}{\partial y^2} \end{bmatrix}$$

① Linear programming

$$f(x, y) = x + y$$

$$\text{s.t. } x + y = 0$$

② Quadratic //

$$f(x, y) = x^2 + y^2$$

$$\text{s.t. } x + y = 0$$

③ non-linear //

$$f(x, y) = x^3 + y^2$$

$$\text{s.t. } x^2 + y = 0$$

$$f(m,s) = 6m^2 + 3s^2$$

M = # hours you study ML per day

S = " " you sleep " "

$$\frac{\partial f(m,s)}{\partial m} = 0 \Rightarrow 12m = 0 \Rightarrow m = 0$$

$$\frac{\partial f(m,s)}{\partial s} = 0 \Rightarrow 0 + 6s = 0 \Rightarrow s = 0$$

$$f(M, S) = 6M^2 + 3S^2 \rightarrow \text{Objective function}$$

$\lambda, \alpha, \beta \leadsto$ Lagrange multiplier

s.t. $M + S = 24$

constraint function

$$g(M, S) = M + S - 24$$

$$L(M, S, \lambda) = f(M, S) - \lambda g(M, S) = 6M^2 + 3S^2 - \lambda (M + S - 24)$$

